

SYNOPSIS

At 0033 on 19 March 2013, the container vessel *CMA CGM Florida* and the bulk carrier *Chou Shan* collided in the East China Sea resulting in both vessels sustaining serious damage, and approximately 610 tonnes of heavy fuel oil being spilled from *CMA CGM Florida*. There were no injuries. *CMA CGM Florida* had left Yang Shan, China, on 18 March and was heading towards Pusan, Korea. *Chou Shan* was heading from Qinhuangdao, China, towards the east coast of Australia.

CMA CGM Florida's Filipino second officer, who was the officer of the watch, altered course to starboard to pass between a group of fishing vessels on the port bow and a vessel on a reciprocal course to starboard. This resulted in a risk of collision with *Chou Shan*, which was crossing *CMA CGM Florida* from port to starboard. *Chou Shan*'s officer of the watch then used the Very High Frequency (VHF) radio to request that *CMA CGM Florida* pass around *Chou Shan*'s stern. The VHF radio conversation was conducted in Mandarin by *CMA CGM Florida*'s Chinese second officer, who had joined the vessel in Yang Shan and was on the bridge for familiarisation.

CMA CGM Florida's Filipino officer of the watch did not understand Mandarin and was unaware that the Chinese second officer had, tacitly, agreed to *Chou Shan*'s request. Both vessels altered course to port, which resulted in a continued risk of collision with each other. *CMA CGM Florida*'s Chinese second officer then called *Chou Shan* on the VHF radio to request that both vessels pass port-to-port. This was agreed to by *Chou Shan*'s officer of the watch. Both vessels then altered course to starboard, resulting in a collision.

CMA CGM Florida's second officers and *Chou Shan*'s officer of the watch considered that it was appropriate to use VHF radio for collision avoidance, contrary to industry advice. Furthermore, *Chou Shan*'s officer of the watch considered that it was appropriate to use VHF radio for negotiating a passing protocol that was contrary to Rule 15 of the International Regulations for Preventing Collisions at Sea.

CMA CGM Florida's Filipino officer of the watch lacked situational awareness. Contributing to this was the Chinese second officer's incomplete translation of the VHF radio communications with *Chou Shan* and the Filipino officer of the watch's disproportionate reliance on Automatic Identification System (AIS) target Closest Point of Approach and Time to Closest Point of Approach information.

CMA CGM International Shipping Company PTE Ltd (CMA CGM) and Sincere Navigation Corporation have each taken action aimed at preventing a recurrence. A recommendation has been made to CMA CGM for it to take appropriate measures to ensure its company shipboard policies and procedures are adhered to and that its masters recognise the importance of engaging with and motivating crew in the safe and efficient management of its vessels. Sincere Navigation Corporation has been recommended to review and amend its safety management system to ensure that VHF radio is not normally used for collision avoidance and that its masters are empowered to provide specific metrics in their standing orders as to when they should be called by the officer of the watch. The International Chamber of Shipping and the Maritime and Coastguard Agency have been recommended to update their respective guidance on the use of AIS data for collision avoidance.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *CMA CGM FLORIDA*, *CHOU SHAN* AND ACCIDENT

SHIP PARTICULARS		
Vessel's name	<i>CMA CGM Florida</i>	<i>Chou Shan</i>
Flag	United Kingdom	Panama
Classification society	Bureau Veritas	American Bureau of Shipping
IMO number/fishing numbers	9348704	9296963
Type	Container vessel	Bulk carrier
Registered owner	Provence Shipowner 2008-1 Limited	Rockwell Shipping Limited
Manager(s)	CMA CGM International Shipping Company PTE Ltd (CMA CGM)	Sincere Navigation Corporation
Construction	Steel	Steel
Keel laid	5 November 2007	1 December 2004
Length overall	294.1m	289.0m
Registered length	Not applicable	Not applicable
Gross tonnage	54,309	91,166
Minimum safe manning	15	14
Authorised cargo	Containers	Bulk cargoes
VOYAGE PARTICULARS		
Port of departure	Yang Shan	Qinhuangdao
Port of arrival	Pusan (intended)	Not specified
Type of voyage	International	International
Cargo information	Containers	Ballast
Manning	24	24

MARINE CASUALTY INFORMATION		
Date and time	19 March 2013 at 0033 (UTC+8)	
Type of marine casualty or incident	Serious Marine Casualty	
Location of incident	140 miles east of Shanghai, East China Sea	
Place on board	Not applicable	Not applicable
Injuries/fatalities	None	None
Damage/environmental impact	Serious damage to port side and accommodation block, Nos 4 and 5 cargo holds flooded, ~610t fuel oil lost	Serious damage to bow area
Ship operation	On passage	On passage
Voyage segment	Mid-water	Mid-water
External & internal environment	Southerly wind force 4, slight sea state, 4 miles visibility, dark	
Persons on board	24	24

1.2 BACKGROUND

1.2.1 *CMA CGM Florida*

CMA CGM Florida was one of 11 vessels serving CMA CGM's Pacific Express (PEX 3) liner service. This service was a 76-day round trip from Hong Kong, which called at 13 ports.

Vessels loaded in Hong Kong, three further ports in China, and Pusan, Korea, before sailing across the Pacific Ocean, transiting the Panama Canal, to various ports on the east coast of the United States of America. They then returned to Hong Kong via Vostochny, Russia. Yang Shan was the fourth port in the rotation. At the time of the accident *CMA CGM Florida* was on passage from Yang Shan towards Pusan at a speed of about 23 knots.

1.2.2 *Chou Shan*

Chou Shan operated on the spot market. When fixed on a charter, the vessel proceeded directly to the loading port in accordance with the charterer's instructions. At the time of the accident, *Chou Shan* was in ballast, on passage from Qinhuangdao, China, towards the east coast of Australia at a speed of about 11.5 knots.

1.3 NARRATIVE

1.3.1 Events leading up to the collision

At about 2340 on 18 March 2013, one of *CMA CGM Florida*'s third officers handed over the officer of the watch (OOV) duty to the Filipino second officer (2/O). The able seaman (AB), acting as lookout, changed at the same time. The traffic situation was discussed during the handover between the officers. However, *Chou Shan*, whose radar target was not displayed, did not feature in this discussion (**Figure 1**).

A Chinese 2/O, who was on board for a period of familiarisation, arrived on the bridge while the handover between the third officer and the Filipino 2/O was underway. Not able to understand the conversation between them as it was conducted in Tagalog, he began to familiarise himself with the bridge paperwork.

After the handover was completed, the Chinese 2/O engaged with the Filipino OOV and began asking questions in English about the bridge equipment and procedures. At times, this took the Filipino OOV away from his position at the port radar to demonstrate equipment operation or the location of documentation.

At 0000 on 19 March 2013, *Chou Shan*'s third officer handed over the watch to the 2/O. The AB engaged as lookout changed at the same time. During the handover, the officers discussed two vessels that were ahead, one to port (*Monte Pascoal*) and one to starboard (*Hong Yun No1*). *CMA CGM Florida*, whose radar target had just appeared on the edge of the radar display, was not discussed (**Figure 2**).

At 0000 on *CMA CGM Florida*, the Global Positioning System (GPS) alarm sounded to indicate the time. This prompted the Filipino OOV to record and plot the vessel's position on the paper chart. Shortly afterwards he looked at the port radar display and observed a number of vessels in the vicinity.

About 1 minute later, the lookout reported, in Tagalog, to the Filipino OOW that he could see fishing vessels on the port bow. The Filipino OOW returned to the port radar and inspected the multiple AIS target list (**Figure 3**).

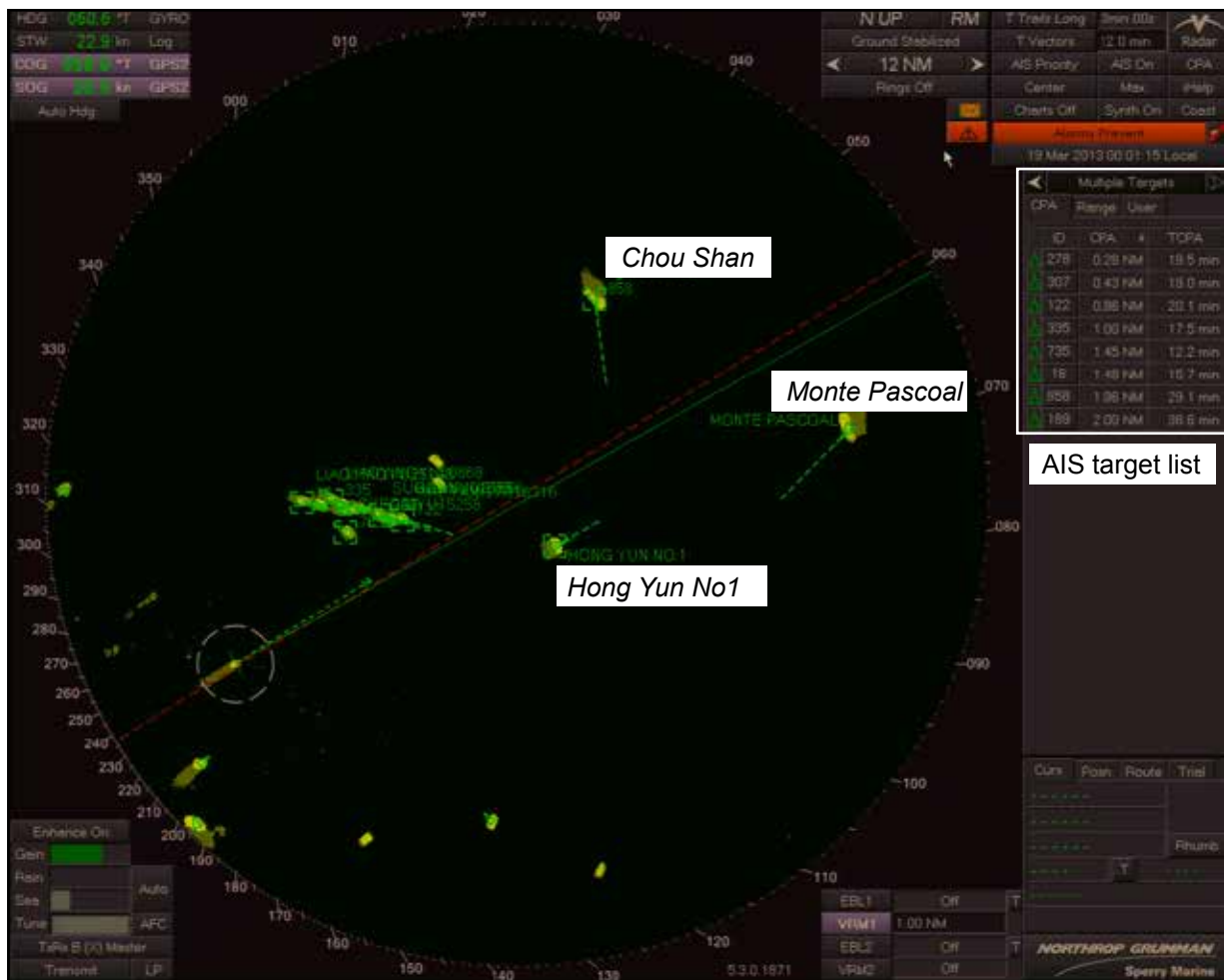


Figure 3: CMA CGM Florida's port radar display at 0002

Monte Pascoal was ahead and to starboard of *CMA CGM Florida* at a range of 17½ miles. The vessel was on an approximately reciprocal course with a Closest Point of Approach (CPA) of more than 2½ miles to starboard.

Hong Yun No1 was also ahead and to starboard of *CMA CGM Florida* at a range of 9 miles. *Hong Yun No1* was steering a similar course to that of *CMA CGM Florida* with a CPA of more than 2 miles to starboard.

Chou Shan was ahead and to port of *CMA CGM Florida* at a range of 14 miles. *Chou Shan* was crossing *CMA CGM Florida*'s track from port to starboard, and had a CPA of more than 1½ miles ahead.

A group of fishing vessels was on the port side of *CMA CGM Florida* at a range of 6 miles. The closest fishing vessel to *CMA CGM Florida*'s bow, *Sheyangyu 16316*, had a CPA of 0.1 mile to starboard. There were two other fishing vessels in the group with a CPA of approximately 0.1 mile.

Chou Shan featured second from the bottom on the multiple AIS target list owing to its relatively large CPA in comparison to that of the fishing vessels. *Hong Yun No1* was at the bottom of the list and *Monte Pascoal* was not listed.

At 0002, the Filipino OOW altered *CMA CGM Florida*'s course by 6 degrees to starboard to 065°(T) using the autopilot. He also reduced the port radar range scale to 6 miles, and selected relative vectors.

At 0007, the Filipino OOW altered *CMA CGM Florida*'s course by a further 5 degrees to starboard, using the autopilot. He also began talking aloud about the fishing vessels, remarking that he did not understand why they did not keep clear and kept changing their course. This was acknowledged, conversationally, by the Chinese 2/O.

The Filipino OOW altered course by 5 degrees to starboard, twice more, to bring *CMA CGM Florida* onto a heading of 080°(T). Just before making the final 5-degree course alteration, *Monte Pascoal* had appeared at the edge of the port radar display.

Throughout the period of course alterations, the Filipino OOW had been alternating the radar range scale between 3, 6 and 12 miles, and had been changing between true and relative vectors/trails.

Becoming concerned about the presence of *Monte Pascoal*, the Filipino OOW asked the Chinese 2/O to call the fishing vessels, in Mandarin, to ask them to keep clear. Using the Very High Frequency (VHF) radio, the Chinese 2/O called *Sheyangyu 16316* but did not receive a response to the call.

Chou Shan's OOW, who was Chinese, manually acquired *CMA CGM Florida*'s radar target to obtain Automatic Radar Plotting Aid (ARPA) information, and then activated the vessel's AIS target to identify it. At this time *CMA CGM Florida*'s CPA was less than ½ mile ahead of *Chou Shan*.

CMA CGM Florida's Filipino OOW again remarked that he did not understand why the fishing vessels did not keep clear, and why "*they wanted to lose their lives*". The Chinese 2/O then made a second VHF radio call to *Sheyangyu 16316*, stating, in Mandarin, that they "*will lose their life*". This second call was not requested by the Filipino OOW.

At 0019, *Chou Shan*'s OOW called *CMA CGM Florida* using his VHF radio. The call was answered by the Chinese 2/O and the resulting conversation was conducted in Mandarin.

The following is a translation of the call and conversation:

Chou Shan: *CMA CGM Florida, CMA CGM Florida. Chou Shan, Chou Shan calling. Over.*

CMA CGM Florida: *Who is calling Florida? Speaking please.*

Chou Shan: *This is Chou Shan ship. Could you pass astern of us please?*

CMA CGM Florida: *I have to avoid that small fishing boat first, and I will turn back after I have avoided it.*

Chou Shan: *Oh, alright. Thank you.*

The following conversations then occurred, in English, between the Filipino OOW and the Chinese 2/O on *CMA CGM Florida*:

Filipino OOW: What does he want?

Chinese 2/O: No... this is another vessel... Chou Shan

Filipino OOW: Chou Shan... ah

Chinese 2/O: Ya... he want we are passing his stern

Filipino OOW: Ah ok

Chinese 2/O: I... I tell him... ah ok... we are passing the fishing boat and we change the course

Filipino OOW: But if you put into the relative they are going to pass behind... only these fishing boats... because... they don't care

Chinese 2/O: Ha

Filipino OOW: Even if you call them ah

Chinese 2/O: Ah you can little bit... er turn back your route... just a little bit

Filipino OOW: So... very close these fishing boats... I don't know

Chinese 2/O: Oh no they change... they change their course

Filipino OOW: Ah they are going to change, they reply they are going to change their course ah

By 0021, *CMA CGM Florida* was drawing ahead of the fishing vessels and the Filipino OOW altered *CMA CGM Florida*'s course 5 degrees to port, to 075°(T).

Chou Shan's OOW continued to monitor *CMA CGM Florida*'s radar target, and commented to the AB that he expected *CMA CGM Florida* to alter course to port once clear of the fishing vessels.

At 0026, *CMA CGM Florida*'s Filipino OOW made a 5-degree alteration of course to port, to 070°(T). He then remarked aloud that *Chou Shan*'s CPA was very small. This prompted the Chinese 2/O to question whether *CMA CGM Florida* would pass *Chou Shan*'s stern. The Filipino OOW stated his expectation that *Chou Shan* would pass *CMA CGM Florida*'s stern, and that both vessels would pass port-to-port.

At the same time, *Chou Shan*'s OOW altered course from 165°(T) to 160°(T), then to 155°(T).

At 0027, the Filipino OOW on *CMA CGM Florida* altered course by 5 degrees to port to 065°(T).

Shortly after making the course alteration, the Filipino OOW stated aloud that, owing to the presence of *Hong Yun No1* ahead, *CMA CGM Florida* and *Chou Shan* should pass port-to-port. The Chinese 2/O then called *Chou Shan*, in Mandarin, on the VHF radio.

Chou Shan was at a range of just under 2 miles, with a CPA of around 0.3 mile ahead (**Figure 4**).

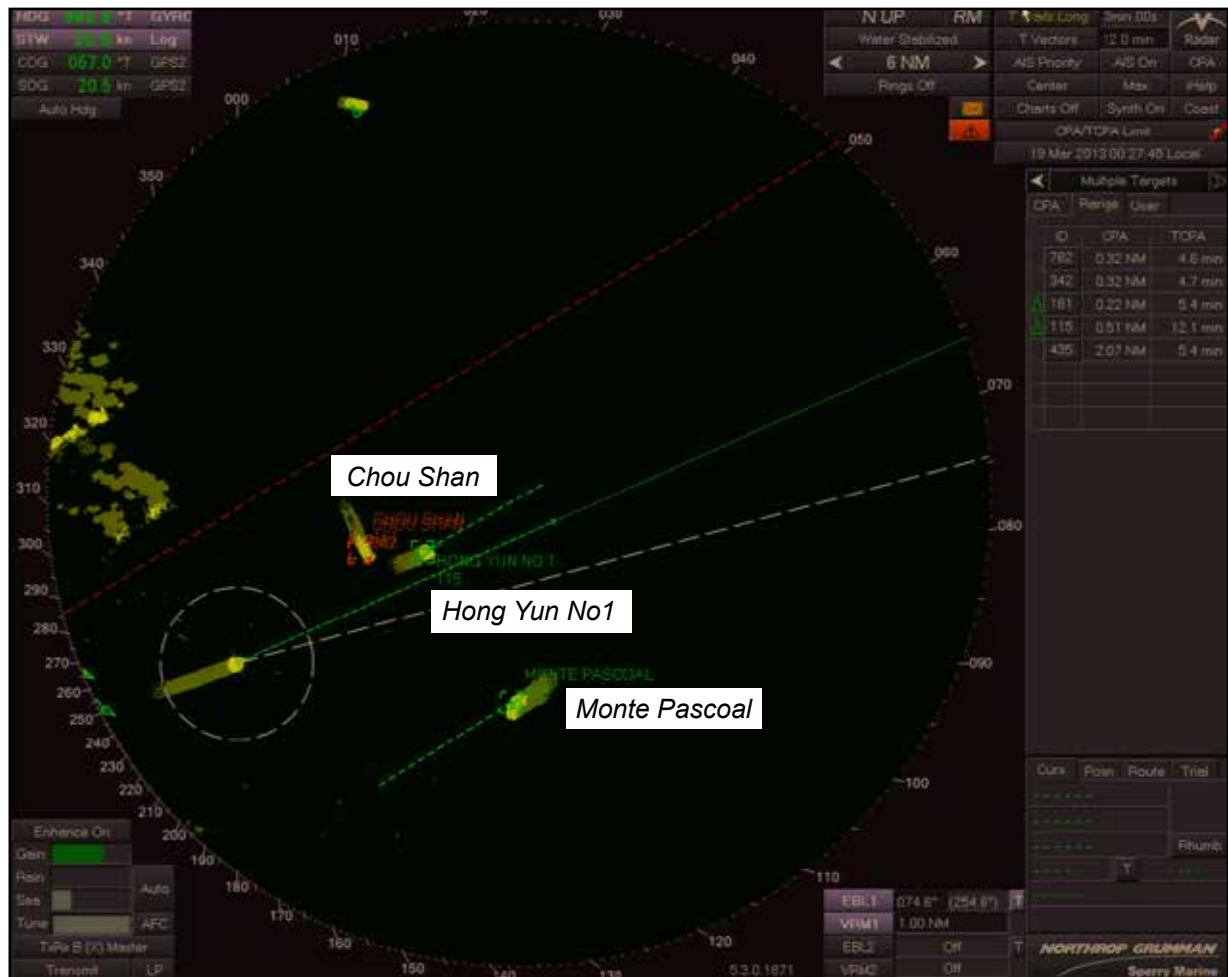


Figure 4: *CMA CGM Florida*'s port radar display at the time of the second VHF radio call to *Chou Shan*

The following is a translation of the call and conversation:

CMA CGM Florida: *Chou Shan, Chou Shan, ... (this is)¹ Florida.*

Chou Shan: *Received. Speaking please.*

CMA CGM Florida: *Ah, Chou Shan, we ... shall we pass each other red to red² shortly please? On my side, in front of me there is (are) still, still fishing boat(s) so that we are unable to pass astern of you, alright?*

¹ Words in brackets have been added by the translator to indicate their interpretation of what was meant.

² Port side to port side.

Chou Shan: Oh? You can pass astern of us. I cannot (agree with what you have suggested) but I can adjust (my direction) a little bit more to the port side.

CMA CGM Florida: Ah, no, that is not possible, not possible. ... our captain wants, wants us to pass each other red to red. Let us pass each other red to red.

Chou Shan: Oh, that means (that I have to make) a large-angle turn.

CMA CGM Florida: Yes. Ah, please turn your ship and we turn ours. Let us pass each other red to red, alright? Many thanks.

Chou Shan: Okay, alright.

CMA CGM Florida: Good, thanks a lot.

Shortly after the VHF radio conversation had finished, *Chou Shan*'s OOW told the AB to take the wheel in manual steering, and ordered 'starboard 20°'. On *CMA CGM Florida*, the Chinese 2/O told the Filipino OOW, in English, that *Chou Shan* had agreed to a port-to-port passing. He provided no further translation of the conversation with *Chou Shan*.

At 0030, concerned that *Chou Shan* did not appear to be altering course, the Filipino OOW instructed the AB to alter *CMA CGM Florida*'s course by 10 degrees to starboard, to 075°(T), using the autopilot.

As *Chou Shan* was turning to starboard, the OOW commented to the AB that he should not have agreed to a port-to-port passing. He then instructed the AB to head towards *CMA CGM Florida*. The AB placed the wheel amidships in preparation to steady the vessel on the required heading, and then suggested 'hard-a-starboard' to the OOW. The OOW agreed and the AB executed the turn at 0031.

CMA CGM Florida's Chinese 2/O continued to question whether *CMA CGM Florida* would pass *Chou Shan*'s stern. The Filipino OOW repeated his expectation that *Chou Shan* would pass *CMA CGM Florida*'s stern, port-to-port, and that he had just altered course to assist with this. He also began flashing the daylight signalling lamp (Aldis lamp), quickly and repeatedly, towards *Chou Shan*.

The Filipino OOW continued to express his doubt as to whether *Chou Shan* had started altering course, and questioned the Chinese 2/O as to whether *Chou Shan* expected a port-to-port passing. The Chinese 2/O said "yes", but suggested that it may be better for *CMA CGM Florida* to pass *Chou Shan*'s stern.

At 0031, the Filipino OOW instructed the AB to take the wheel and ordered 'hard-a-port'. Shortly afterwards, he ordered 'steady' and then 'hard-a-starboard'.

As both vessels continued to close each other, the Filipino OOW continued to flash the daylight signalling lamp before calling the master about 18 seconds before the collision. The Filipino OOW reported to the master that *CMA CGM Florida* was very close to another vessel (**Figures 5 and 6**). *Chou Shan*'s master was not called prior to the collision.

1.3.2 Collision

At 0033, *Chou Shan*'s bow hit *CMA CGM Florida*'s port side. *Chou Shan* had approached on a heading about 40 degrees from *CMA CGM Florida*'s fore and aft line, and struck in the vicinity of No.5 cargo hold immediately forward of the accommodation block.

1.3.3 *CMA CGM Florida* - immediate post-collision actions

The general alarm was sounded on *CMA CGM Florida* as the master stepped through the bridge doorway. He asked the Filipino OOW what had happened, ordered the main engine to be stopped, observed *Chou Shan* astern, and asked for the collision checklist.

The chief engineer proceeded to his muster station in the engine control room, and then went into the engine room to inspect for damage. He reported to the bridge that the engine room was free of damage. However, he was concerned that the levels in the engine room port low sulphur fuel oil (LSFO) tank, and No.5 upper high sulphur fuel oil (HSFO) tank had dropped, and so went to the main deck to investigate.

Meanwhile, the master told the chief officer to use the vessel's heeling system to correct the list that had developed. All other crew members proceeded to their muster station. However, no report of who was present was made to the bridge. Later, one of the third officers was instructed to go to the bridge and help to maintain a contemporaneous log of events.

Recalling the vessel's departure condition, the master ordered that No.6 ballast tank be pumped out to reduce the stresses on the hull. At about the same time, the chief engineer was attempting to transfer oil from what he now believed to be breached fuel oil tanks. This was largely unsuccessful as the valve control station, situated in the port underdeck passage, was destroyed in the collision. The master then ordered no further ballast transfers until the vessel's stability had been verified.

The master then called CMA CGM's emergency telephone number to report the accident. He then contacted *Chou Shan* using the VHF radio. Previous calls from *Chou Shan* had not been answered.

At about 0100, all crew were accounted for, and a systematic inspection of the vessel began to establish the extent of damage. Individual crew members were ordered to carry out inspections of particular areas, according to the master's instructions. Contact had now been made with the Bureau Veritas damage stability assessment service, and it was making an assessment of the vessel's condition.

By 0310, a full inspection of the vessel had been completed, and all tanks had been sounded. Being concerned about the extent of the damage sustained, the master instructed all crew members, except the chief engineer, chief officer and one of the third officers, to muster in the crew mess room with lifejackets and immersion suits.

At 0644, a telephone call was made by CMA CGM to the Maritime Rescue Co-ordination Centre (MRCC) Falmouth to report the accident. This report was forwarded by MRCC Falmouth to MRCC Shanghai.

1.3.4 Post-collision events

Following the collision, *Chou Shan*'s crew were mustered and an inspection of the vessel's forward part carried out. Significant damage was reported, however it was found to be limited to the area forward of the collision bulkhead. *Chou Shan*'s master reported the accident to the vessel's manager and regular soundings of appropriate spaces continued overnight.

Chou Shan proceeded to dry dock during the afternoon of 19 March. Due to the resulting oil pollution caused by ruptured fuel oil tanks, *CMA CGM Florida* was required to await inspection before any repair arrangements could be made.

A salvage vessel was contracted and officials from China Maritime Safety Administration boarded *CMA CGM Florida* to carry out an inspection. On 21 March, the vessel was anchored 80 miles off Shanghai so that the damaged tanks could be cleaned prior to allowing the vessel to proceed into port.

CMA CGM Florida received clearance to proceed towards Yang Shan on 26 March, and berthed alongside on 27 March. After a naval architect was consulted, most of the containers were discharged from the vessel, and the vessel then proceeded to dry dock for repairs.

1.4 DAMAGE AND POLLUTION

1.4.1 *CMA CGM Florida*

CMA CGM Florida suffered extensive damage to the port side (**Figures 7 and 8**). Damage was focused in the vicinity of No.5 cargo hold and the outboard No.5 upper HSFO tank, which were holed above and below the waterline. No.4 cargo hold and the engine room LSFO tank adjacent to No.5 cargo hold were also breached.

Above the main deck, the accommodation block was damaged over five decks, and the port lifeboat was seriously damaged, rendering it unusable. Five pontoon-type hatch covers were damaged and 263 containers were damaged or lost overboard.

No.s 4 and 5 cargo holds flooded as a result of the collision. Breach of the engine room LSFO tank and No.5 upper HSFO tank resulted in the loss of around 610t of fuel oil.

There were no injuries and the resulting oil pollution was contained and cleaned up with minimal environmental impact.

1.4.2 *Chou Shan*

Chou Shan sustained serious damage to its bow, limited to the area forward of the collision bulkhead (**Figures 9 and 10**). The full force of the impact was concentrated on its port side due to the angle of approach with *CMA CGM Florida*.

The fore peak tank was opened to the sea. No other compartments were breached as a result of the collision. Considerable shell plate and frame damage occurred to the bow area. The port anchor and hawse pipe were destroyed.

There were no injuries and no oil pollution resulting from *Chou Shan*'s damage.

1.5 ENVIRONMENTAL CONDITIONS

At the time of the accident, the wind was southerly force 4, and the sea state was slight with a low swell. It was dark with haze. There were various reports regarding the range of visibility, the lowest being 4 miles.

1.6 MANNING AND WATCHKEEPING

1.6.1 CMA CGM Florida

Overview

CMA CGM Florida had a safe manning document³ requiring a minimum of 15 crew members. At the time of the accident there were 24 crew members on board. This total included two trainees and two officers undergoing familiarisation.

The senior officers (i.e. master, chief officer, chief engineer and second engineer) were Romanian nationals. Junior officers and ratings were predominantly from the Philippines.

The working language on *CMA CGM Florida* was English.

Key personnel

- Master

The master was 38 years old. He held a STCW⁴ II/2 Master Unlimited Certificate of Competency (CoC), which had been issued in 2008 by the Government of Romania. He had also been issued an appropriate Certificate of Equivalent Competency (CEC) by the United Kingdom's (UK) Maritime and Coastguard Agency (MCA).

He had worked at sea for 15 years, having joined CMA CGM in 1998. In 2008, he was promoted to master and served successive contracts on different vessels within the fleet.

The master had joined *CMA CGM Florida* for the first time on 3 February 2013 on a normal 4-month contract. Since joining, he had not discussed with the deck officers any detailed requirements with regard to bridge watchkeeping.

- Filipino 2/O

The Filipino 2/O was 50 years old. He held an STCW II/1 OOW Unlimited Certificate, which had been issued by the Government of the Republic of the Philippines in 2004. He had also been issued an appropriate CEC by the MCA.

He graduated with a degree in marine transportation in 1982 and, after completing his deck apprenticeship in 1984, worked ashore. From 1991 he worked at sea on refrigerated cargo vessels and tankers as an AB.

³ Document issued by a vessel's Flag State detailing the minimum number of crew, and their qualifications, required to be on board to operate the vessel safely.

⁴ International Convention on Standards of Training, Certification and Watchkeeping for Seafarers

1.10 VOYAGE DATA RECORDER RECOVERY AND DATA

1.10.1 CMA CGM Florida

Good quality video and audio records were successfully retrieved from *CMA CGM Florida*'s Sperry Voyage Data Recorder (VDR). All required sensor inputs were in place and had recorded correctly.

1.10.2 Chou Shan

Chou Shan's JRC VDR data was downloaded about 48 hours after the accident by a technician. Despite having all the required sensor inputs in place, exactly 1 minute of audio data was found absent from all voice sensors. The missing minute of audio data coincided with *Chou Shan*'s OOW's request for *CMA CGM Florida* to pass astern of *Chou Shan*. Fortunately, the VHF radio conversation at this time was recorded fully on *CMA CGM Florida*'s VDR. The reason why this data was lost has not been established.

1.11 SIMILAR ACCIDENTS

1.11.1 Recent MAIB investigations

Spring Bok/Gas Arctic (MAIB report No 24/2012)

On 24 March 2012, the Netherlands-registered cargo vessel *Spring Bok* collided with the Maltese-registered liquefied petroleum gas tanker *Gas Arctic*. The collision occurred in less than 2 miles visibility, 6 miles south of Dungeness while the vessels were proceeding in the same direction in the south-west lane of the Dover Strait Traffic Separation Scheme. There were no injuries or pollution, but both vessels suffered structural damage.

The MAIB investigation concluded that, despite each vessel detecting and identifying the other by radar and AIS, neither OOW made a full appraisal of the risk of collision, nor took the action required by the COLREGS to prevent the collision.

Scottish Viking/Homeland (MAIB Report No 4/2010)

On 5 August 2010, the Italian-registered Ro-Ro passenger ferry *Scottish Viking* collided with the fishing vessel *Homeland* about 4 miles off St Abbs Head, UK. As a result of the collision, the fishing vessel sank and one of her crew was lost.

The MAIB investigation concluded that *Scottish Viking*'s OOW did not: determine at an early stage if there was a risk of collision with *Homeland*; sufficiently monitor or plot *Homeland*'s track; and, once a risk of collision was deemed to exist, take sufficient action to avoid a collision.

1.11.2 Request to the Marine Accident Investigators' International Forum (MAIIF)

As part of its investigation, the MAIB requested MAIIF members for information on marine casualties or incidents featuring the following:

1. Lack of foresight and substantial early action to avoid a developing congested traffic situation.

2. Significant reliance on AIS information to determine risk of collision solely in terms of CPA.
3. Use of VHF radio communication to negotiate action to avoid collision.
4. Agreement to take avoiding action in conflict with action required by the COLREGS.

Table 1 below summarises the responses received from MAIIF members. The MAIB received a total of seven responses from MAIIF members, referring to eight marine casualties or incidents featuring one or more of the specified safety issues.

Safety issue	Number of marine casualties or incidents
Lack of foresight and substantial early action to avoid a developing congested traffic situation.	5
Significant reliance on AIS information to determine risk of collision solely in terms of CPA.	0
Use of VHF radio communication to negotiate action to avoid collision.	5
Agreement to take action in conflict with action required by the COLREGS.	0

Table 1: Summary of responses received from MAIIF members

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 FATIGUE

There is no evidence that any of the crew involved in the accident were suffering from fatigue. Therefore, fatigue is not considered to be a contributing factor to this accident.

2.3 OVERVIEW

CMA CGM Florida's OOW, the Filipino 2/O, initially altered course to pass between the fishing vessels and *Monte Pascoal*, which resulted in a risk of collision with *Chou Shan*. *Chou Shan's* OOW used the VHF radio to call *CMA CGM Florida* to request it to pass around the stern of *Chou Shan*. The Chinese 2/O on *CMA CGM Florida*, who was undergoing familiarisation, had tacitly agreed to this course of action.

Both *CMA CGM Florida* and *Chou Shan* altered course to port following the VHF radio agreement, which resulted in a continued risk of collision with each other.

CMA CGM Florida's Chinese 2/O then called *Chou Shan* on the VHF radio to request that both vessels pass port-to-port. This was agreed to by *Chou Shan's* OOW. Both *Chou Shan* and *CMA CGM Florida* altered course to starboard, but these measures were insufficient and taken too late to prevent collision.

2.4 ACTION BY *CMA CGM FLORIDA'S* FILIPINO OOW

2.4.1 Situational assessment at 0002

At 0002, the Filipino OOW made the following assessments of the risk of collision with vessels in the vicinity of *CMA CGM Florida* and his obligations under the COLREGS. The assessments were primarily based on the radar-displayed AIS target CPA information and his general overview of the displayed radar targets. Three merchant vessels and one group of fishing vessels featured in these assessments.

The Filipino OOW correctly interpreted that, in accordance with Rule 13 of the COLREGS, *CMA CGM Florida* was overtaking *Hong Yun No1* and was required to keep out of her way.

The Filipino OOW also correctly interpreted that there was a risk of collision with the fishing vessels, and these became the focus of his attention. It is unclear what navigation lights the fishing vessels were exhibiting. Regardless of whether they were those of vessels engaged in fishing or power-driven vessels underway, the Filipino OOW correctly interpreted that *CMA CGM Florida* had an obligation to keep out of the way under Rule 13 of the COLREGS.

2.4.2 Action to avoid fishing vessels

The Filipino OOW's initial starboard course alteration of only 5 degrees, followed by three further 5-degree starboard course alterations, was inappropriate in that they would not have been readily apparent to other vessels, contrary to Rule 8(b). Furthermore, the initial 5-degree alteration of course did not constitute substantial action, contrary to Rule 16.

The Filipino OOW's decision to alter course in 5-degree steps resulted from a lack of situational awareness in that he had not determined the course and speed of the fishing vessels and so could not assess what course alteration would be necessary for *CMA CGM Florida* to pass safely ahead of them. Instead of using ARPA to ascertain radar target compass bearing, course and speed, and to conduct a radar target trial manoeuvre, he continued predominantly to use the multiple AIS target list from which he was able to ascertain AIS target CPA and TCPA only. Consequently, in an attempt to restrict the overall course alteration to a minimum and not wishing to impede the passage of *Monte Pascoal*, he conducted a series of small reactive course alterations rather than a single and substantial proactive one.

The Filipino OOW's initial course alteration to starboard coincided with his reducing the port radar display range scale to 6 miles, on which it predominantly remained. This meant that only the radar targets of the fishing vessels and *Hong Yun No1* were displayed. *Chou Shan* featured second from bottom on the multiple AIS target list as fishing vessels with smaller CPAs featuring at the top of the list. *Monte Pascoal*'s radar target appeared on the display following the third 5-degree course alteration. Therefore, in making the final 5-degree course alteration, the Filipino OOW would have been aware that *CMA CGM Florida* was now showing a port aspect to *Monte Pascoal*.

Precautionary thought (ie the ability to analyse the effect of an intended manoeuvre) is an important skill that every OOW should have and reflects the requirements of Rule 2(a). Instead of deciding to make an initial course alteration of 5 degrees to starboard, the Filipino OOW could have made a far greater starboard course alteration, taking *CMA CGM Florida* well clear of the bottleneck that was developing ahead.

Before taking action to avoid the vessels posing the most immediate risk of collision, it would have been appropriate for the Filipino OOW to assess the movements of vessels at greater range to ensure that by resolving one risk of collision he was not creating another.

2.4.3 VHF radio communications

After altering course for the fishing vessels, interpreting that a risk of collision with them still existed but not wanting to alter course further to starboard, the Filipino OOW asked the Chinese 2/O to call the fishing vessels on VHF radio and request that they keep clear of *CMA CGM Florida*.

He asked the Chinese 2/O to do this because he believed there would be a better chance of a positive outcome from the VHF radio call if it was made in Mandarin, which he assessed to be the likely first language of the fishing vessel crews.

Both 2/Os considered that it was appropriate to use the VHF radio in collision avoidance situations having identified the names of the vessels concerned using the AIS.

This request had three consequences. The first was that it made the Chinese 2/O an active member of the bridge team. Secondly, it effectively announced to other vessels in the vicinity that a Mandarin-speaking watchkeeper was on *CMA CGM Florida*'s bridge, and thirdly, it indicated that there was a willingness to negotiate passing protocols for collision avoidance. Specifically, it was hearing the VHF radio call transmitted in Mandarin that prompted *Chou Shan*'s OOW to call *CMA CGM Florida*.

As the VHF radio call took place in Mandarin, the Filipino OOW was not aware of its content and he relied on the Chinese 2/O's feedback to him in English. He was still focused on the fishing vessels and *Monte Pascoal*, initially thinking that the VHF radio call had, in fact, come from *Monte Pascoal*.

The fact that *Chou Shan* had requested *CMA CGM Florida* to pass astern was correctly conveyed. However, the fact that a tacit agreement had been made by the Chinese 2/O was not passed on to the Filipino OOW.

When the Filipino OOW checked the relative vector of *Chou Shan*'s radar target, it showed the vessel passing astern of *CMA CGM Florida*. His assessment of the situation was that if a risk of collision existed, *Chou Shan* would be the give-way vessel and would avoid passing ahead in accordance with Rule 15 of the COLREGS. He therefore dismissed *Chou Shan*'s request for *CMA CGM Florida* to pass astern of *Chou Shan* as being inappropriate.

However, being unaware that the Chinese 2/O had tacitly agreed to the course of action, he made no attempt to reply to *Chou Shan*'s request. This left the watchkeeping officers on both vessels with different expectations about what the other vessel would do.

When *CMA CGM Florida*'s Chinese 2/O later called *Chou Shan* in Mandarin and agreed a port-to-port passing, he omitted to inform the Filipino OOW in English that he had also agreed that *CMA CGM Florida* would alter course to starboard to assist the passing. He also omitted to say that this agreement had been made in response to *Chou Shan*'s OOW's concern over the large turn that his vessel would have to make. These omissions again left the Filipino OOW with an erroneous expectation as to what the other vessel would do.

2.4.4 Traffic situation and course alteration after passing the fishing vessels

By 0021:

- *CMA CGM Florida* was drawing ahead of the fishing vessels.
- *Monte Pascoal* was on the starboard bow of *CMA CGM Florida* at a range of 6 miles and with a CPA of 2.4 miles to starboard.
- *Hong Yun No1* was ahead and to port of *CMA CGM Florida* at a range of 4 miles with a CPA of 1.1 miles to port.

- *Chou Shan* was ahead and to port of *CMA CGM Florida* at a range of 4½ miles with a CPA of 0.2 mile astern.

As an overtaking vessel, *CMA CGM Florida* still had an obligation to keep out of the way of *Hong Yun No1*. A risk of collision now existed with respect to *Chou Shan* and, as a stand-on vessel in a crossing situation, *CMA CGM Florida* was required to maintain its course and speed in accordance with Rule 17(a)(i) of the COLREGS. There was no requirement for the Filipino OOW to take any action with respect to *Monte Pascoal*.

The Filipino OOW's primary concern was to open up sufficient sea room with *Monte Pascoal* so as not to impede its passage. Although aware of the presence of *Chou Shan*, he was comfortable in his assessment that *Chou Shan*'s OOW would take avoiding action in the event of risk of collision.

The Filipino OOW then made three 5-degree course alterations to port. This action had the desired effect of increasing *Monte Pascoal*'s CPA and reducing *CMA CGM Florida*'s deviation from the planned track. However, it was contrary to Rules 17(a)(i) and (c), did not remove the risk of collision with *Chou Shan*, and reinforced in the mind of *Chou Shan*'s OOW that *CMA CGM Florida* would alter course around his vessel's stern, as he had earlier requested.

2.4.5 Action to avoid *Chou Shan*

At 0026, with *CMA CGM Florida* on a heading of 070°(T), *Chou Shan* was ahead and to port at a range of 2.9 miles and with a CPA of 0.2 mile ahead. The Filipino OOW's attention was now focused on *Chou Shan* and he stated his concern over that vessel's small CPA to the Chinese 2/O. Despite the Chinese 2/O then conveying his expectation that *CMA CGM Florida* would pass around *Chou Shan*'s stern, the Filipino OOW remained resolved in his opinion that *Chou Shan* should pass around *CMA CGM Florida*'s stern. At 0027, the Filipino OOW altered *CMA CGM Florida*'s course to port by a further 5 degrees and stated aloud to the Chinese 2/O that the vessels should pass port-to-port.

Although initially required to maintain course and speed in accordance with Rule 17(a)(i) of the COLREGS, the Filipino OOW had an option of taking avoiding action himself, in accordance with Rule 17(a)(ii), as soon as he considered that *Chou Shan* was not taking appropriate avoiding action. As *Monte Pascoal* was now drawing clear, appropriate precautionary actions would have been for the Filipino OOW to have sounded at least five short and rapid blasts on the whistle supplemented by a light signal of at least five short and rapid flashes, in accordance with Rule 34(d), and to have then altered course to starboard. However, although concerned, the Filipino OOW was not yet in doubt that *Chou Shan* would take appropriate action.

Chou Shan was under 2 miles away on the port bow, steering a course at right angles to that of *CMA CGM Florida*. The Filipino OOW lacked situational awareness in that he underestimated the time it would take for *Chou Shan* to complete its turn so as to pass port-to-port with *CMA CGM Florida*. This was because he lacked an appreciation of *Chou Shan*'s likely rate of turn and/or how large a turn *Chou Shan* would have to make, given its relative bearing, range and heading at the start of the turn. A number of factors potentially contributed to his lack of appreciation:

- The Filipino OOW may not have appreciated the significance of the aspect presented by the navigation lights of *Chou Shan*.
- In the period leading up to this point, *Chou Shan*'s displayed radar target vector and trail were predominantly relative.
- Use of the multiple AIS target list meant that *Chou Shan*'s course and speed information was not displayed.
- Although *Chou Shan*'s OOW had indicated to *CMA CGM Florida*'s Chinese 2/O that he was concerned over the large turn he would have to make, this information had not been passed to the Filipino OOW.

At 0030, seeing no apparent change to *Chou Shan*'s visual aspect or CPA, the Filipino OOW began to doubt that *Chou Shan* was taking appropriate avoiding action. He decided to alter *CMA CGM Florida*'s course by 10 degrees to starboard and to repeatedly flash the daylight signalling lamp in the direction of *Chou Shan*. Although this action was in the spirit of Rules 17(a)(ii) and 34(d), the success of *Chou Shan*'s manoeuvre now required a significant alteration of course by *CMA CGM Florida* in accordance with Rule 17(b). This occurred about 1 minute later, when the Filipino OOW ordered 'hard-a-starboard' and then called the master. However, his avoiding action was too late to prevent the collision.

2.5 ACTION BY CMA CGM FLORIDA'S CHINESE 2/O

The Chinese 2/O was on the bridge to receive a handover from the Filipino OOW and to familiarise himself with the bridge equipment. Neither officer had received any guidance as to how the handover and familiarisation should be conducted during the watch. Therefore, when they arrived on the bridge, each had individual expectations and priorities.

The Chinese 2/O realised that this would be his main opportunity to learn about the bridge equipment before taking on the duty of OOW at sea, and he was, therefore, focused on his familiarisation. The Filipino OOW's focus was on safely discharging his duties as OOW.

The Chinese 2/O did not have a defined role within the bridge team. However, he was willing to temporarily join the bridge team to assist as required before reverting back to his familiarisation role.

When he was asked by the Filipino OOW to call the fishing vessels on the VHF radio, he readily accepted the task believing his ability to communicate externally, in the local language, to be helpful in the circumstances. Use of the VHF radio for collision avoidance was common practice in his experience.

By accepting this task, the Chinese 2/O had assumed an external communicator role within the bridge team. As the Filipino OOW did not understand Mandarin, the success of this role relied heavily on accurate transmission of the Filipino OOW's intentions, with correct and comprehensive translation of any received responses.

Not otherwise engaged with the conduct of the watch, the Chinese 2/O was not fully aware of the navigational situation or the potential risks to *CMA CGM Florida* posed by other vessels in the vicinity. Therefore, when the two officers began discussing the fishing vessel movements after the first VHF radio call, he was unable to provide any informed opinion on the navigational situation.

The second VHF radio call was not requested by the Filipino OOW. The Chinese 2/O autonomously transmitted the call, again in Mandarin, in his assumed role as external communicator, believing this to be helpful to the Filipino OOW. By not questioning the content of the VHF radio call or the Chinese 2/O's autonomous action, the Filipino OOW effectively gave tacit approval for the Chinese 2/O to continue to act in this way.

Therefore, when *Chou Shan's* OOW called on the VHF radio, the Chinese 2/O autonomously responded. Having answered the call, he was now drawn into a VHF radio conversation with *Chou Shan's* OOW, in Mandarin, during which he, in effect, agreed that *CMA CGM Florida* would alter course to pass *Chou Shan's* stern, without first consulting the Filipino OOW or fully translating the conversation into English for him.

At this time, the Chinese 2/O had noticed that *CMA CGM Florida* was south of her intended track, and assumed that the Filipino OOW would alter *CMA CGM Florida's* course to port to regain the planned track after passing clear of the fishing vessels.

When the Chinese 2/O translated the VHF radio conversation to the Filipino OOW, he did not convey that he had tacitly agreed that *CMA CGM Florida* would pass astern of *Chou Shan*. Instead, he urged the Filipino OOW to alter *CMA CGM Florida's* course to port. He also tried to convince him that the fishing vessels would change their course, the validity of which was challenged by the Filipino OOW.

This challenge caused the Chinese 2/O to make a further unanswered VHF radio call to *Sheyangyu 16316*, asking the fishing vessel's skipper to alter course. The Chinese 2/O then reverted to familiarising himself with the navigation equipment and he neither questioned the Filipino OOW's understanding of what he had agreed nor monitored the situation with *Chou Shan*. He did not consider these tasks to be necessary as they were not his responsibility.

However, the Chinese 2/O demonstrated a continued willingness to offer his opinion when the Filipino OOW voiced his concern regarding *Chou Shan's* small CPA. This re-opened their earlier, unresolved conversation, and he again suggested that *CMA CGM Florida* should pass *Chou Shan's* stern.

The Chinese 2/O's expectation was in direct conflict with the Filipino OOW's interpretation and intention with regard to *Chou Shan*. This was a confusing situation, which almost certainly contributed to the Filipino OOW's delay in asserting his requirement that *CMA CGM Florida* and *Chou Shan* should pass port-to-port. Again, this challenge caused the Chinese 2/O to revert to his external communicator role, and he then called *Chou Shan* on the VHF radio to request a port-to-port passing.

During the VHF radio conversation, *Chou Shan's* OOW expressed concern over his vessel's ability to make a large enough turn to enable a port-to-port passing. Tasked with getting agreement, and possibly fearing loss of face, *CMA CGM Florida's*

Chinese 2/O leveraged his national cultural power-distance hierarchy⁸ by stating that his “captain” wanted the two vessels to pass in this way. This influenced *Chou Shan*’s OOW to re-negotiate, and then to agree to the port-to-port passing following the Chinese 2/O’s stated intention for *CMA CGM Florida* to also alter course.

After the VHF radio conversation, the Chinese 2/O informed the Filipino OOW that *Chou Shan* had agreed to pass port-to-port, but not that he had stated that *CMA CGM Florida* also would alter course.

Having obtained the agreement, the Chinese 2/O again adopted a passive role within the bridge team. However, he considered that *CMA CGM Florida* could still pass astern of *Chou Shan*, and offered this opinion to the Filipino OOW in the short time leading up to the collision. This would have added further confusion to the situation and was inappropriate given that the Chinese 2/O was fully aware that *Chou Shan* would be turning to starboard. Any turn to port at this time had the potential to take *CMA CGM Florida* across the path of *Chou Shan*, contrary to Rule 17(c) of the COLREGS, and would have been highly dangerous.

Summary

By inviting the Chinese 2/O to communicate in Mandarin on VHF radio on his behalf, the Filipino OOW unnecessarily put himself in the position of having to deal with the consequences of the Chinese 2/O’s VHF radio conversations, which he did not understand and was unable to control. He was now reliant on the Chinese 2/O accurately transmitting his intentions, and translating his received responses correctly and comprehensively into English.

The Chinese 2/O did not convey fully to the Filipino OOW the content and outcome of his VHF radio communications with *Chou Shan*’s OOW. Whether or not the two individuals had direct interpersonal conflicts is unclear. However, they were both hampered in their attempts to communicate by having to converse in a second language. Current research suggests that culturally, the Chinese 2/O, although of equivalent rank to that of the Filipino OOW, is likely to have been respectful of the Filipino OOW’s age, experience and authority as the OOW⁸. Research has also identified industry concerns about the compatibility of Chinese and Filipino nationals on board some vessels⁹.

2.6 ACTIONS BY CHOU SHAN’S OOW

2.6.1 Initial action to avoid collision with *CMA CGM Florida*

By 0019, *CMA CGM Florida* had altered course to 080°(T), resulting in a risk of collision with *Chou Shan*. Having previously heard the VHF radio call from *CMA CGM Florida* to *Sheyangyu 16316*, and then having identified *CMA CGM Florida*’s radar target by AIS and having obtained full radar target information for her, the OOW called *CMA CGM Florida* on VHF radio and requested that the vessel pass astern of *Chou Shan*.

⁸ Referred to in “Effects of national culture on human failures in container shipping: The moderating role of Confucian dynamism” (reproduced in Annex C), Sections 2.1, 2.2, 2.5, 4.1, 4.2 and 5.1

⁹ “The Impact of Multicultural and Multilingual Crews on Maritime Communication” - The MARCOM Project (1999). Pages 69-70 of this report provide details of a questionnaire provided to a number of shipping companies in 1996. Responses from a number of these companies to a question ‘if the respondents felt that, in their experience, there were mixes of nationalities that were not compatible on ships’, placed at number one on the list, Chinese and Filipino incompatibility.

Chou Shan was a give-way vessel in a crossing situation and was required to avoid crossing ahead of *CMA CGM Florida* in accordance with Rule 15 of the COLREGS. However, on hearing the VHF radio call transmitted in Mandarin from *CMA CGM Florida*, *Chou Shan*'s OOW was alerted to the fact that a Mandarin-speaking watchkeeper was on her bridge and that there was an opportunity to negotiate a passing protocol for the crossing vessels.

In the absence of evidence from which to conclude an alternative explanation, a VHF radio call to *CMA CGM Florida* to request the vessel to pass astern of *Chou Shan*, albeit contrary to Rule 15 of the COLREGS, presented the OOW with a convenient solution to the risk of collision that dispensed with the need for *Chou Shan* to take avoiding action.

The OOW interpreted *CMA CGM Florida*'s subsequent course alterations to port on clearing ahead of the fishing vessels as action in accordance with the agreed manoeuvre. Despite the resulting minimal CPA, no agreement had been made as to when and by how much *CMA CGM Florida* would alter course and so the OOW had no reason to doubt that things were going to plan. However, by 0026, he had become concerned that *CMA CGM Florida* was not taking sufficient action to safely pass astern and altered *Chou Shan*'s course by 10 degrees to port. This had the effect of increasing *CMA CGM Florida*'s TCPA. At that point, it would also have been appropriate for the OOW to have indicated his concern to *CMA CGM Florida* by sounding at least five short and rapid blasts on the whistle supplemented by a light signal of at least five short and rapid flashes.

2.6.2 *CMA CGM Florida*'s request to pass port-to-port

When he received the request for a port-to-port passing from *CMA CGM Florida*, the OOW initially challenged it, demonstrating that he was aware how difficult it would be for *Chou Shan* to make a turn to starboard at that stage to leave *CMA CGM Florida* clear on *Chou Shan*'s port side.

The statement received on VHF radio that *CMA CGM Florida*'s master required a port-to-port passing clearly had an effect on the OOW's decision-making. This was probably due, at least in part, to the high power-distance hierarchy of his national culture and the consequent level of respect he would ordinarily give to a master. His concerns about the proposed manoeuvre were further allayed by the stated intention that *CMA CGM Florida* would also alter course. The result was that, despite his reservations, he agreed to a port-to-port passing.

Having allowed *Chou Shan* to approach within 2 miles of *CMA CGM Florida*, becoming concerned about that vessel's inaction following the initial course alterations to port, and now faced with this request for a port-to-port passing about which he was clearly in doubt, the OOW should by now have called the master. His reason for not doing so at any time prior to the collision has not been established from the available evidence.

Sincere Navigation Corporation's SMS and *Chou Shan*'s master's standing orders required an OOW to notify the master when traffic conditions or movements of other vessels caused concern. This presumed that the OOWs and the master had a mutual understanding of the circumstances in which the OOWs would become sufficiently concerned to warrant their calling the master.

2.8 USE OF VHF RADIO FOR COLLISION AVOIDANCE

Following each of the two VHF radio communications between *CMA CGM Florida* and *Chou Shan*, the Filipino OOW and *Chou Shan*'s OOW were left with different expectations. A significant contributing factor to this misunderstanding was that the communication was conducted in a language which the Filipino OOW was unable to understand.

CMA CGM Florida's two 2/Os and *Chou Shan*'s OOW considered that it was appropriate to use VHF radio for collision avoidance. The latter also considered that it was appropriate to use it for negotiating a manoeuvre that was contrary to Rule 15 of the COLREGS.

The ICS's Bridge Procedures Guide recommends against using VHF radio for collision avoidance and warns that, even where vessels have identified each other, misunderstandings may still arise. This warning is reflected in the MCA's MGN 324 (M+F), which further advises that even when VHF radio is resorted to, the COLREGS should be complied with.

In an internal feedback publication, CMA CGM had previously indicated that use of VHF radio to avoid collision was not recommended. However, its SMS provided no specific instructions regarding the use of VHF radio for collision avoidance and made no mention of this in its general guidance as to what the master's standing orders should include. This left the management company's expectations on the use of VHF radio for collision avoidance open to interpretation by its masters. *CMA CGM Florida*'s master's standing orders were ambiguous and did not fully reflect the above recommended practice.

The instructions provided in Sincere Navigation Corporation's SMS were in direct conflict with the guidance contained in the ICS's Bridge Procedure Guide.

The use of VHF radio for collision avoidance was unnecessary, was contrary to internationally recognised best practice, and was a significant contributing factor to the collision.

2.9 USE OF AIS AND RADAR FOR COLLISION AVOIDANCE

2.9.1 CMA CGM Florida

CMA CGM Florida's Filipino OOW lacked situational awareness. Consequently, he altered his vessel's course in 5-degree steps to avoid the fishing vessels being overtaken, and gave no consideration to making a larger, more strategic alteration of course to starboard at an early stage to avoid vessels further ahead. A contributing factor to his lack of situational awareness was his use of AIS and radar for collision avoidance. His risk of collision assessments were primarily based on the radar-displayed AIS target CPA information and his general overview of the displayed radar targets.

The Filipino OOW used AIS priority and the multiple AIS target list, which displayed CPA and TCPA in ascending order either of CPA or range. This encouraged him to focus on those vessels with the smallest CPA or range at the expense of maintaining a more strategic overview of the traffic situation.

Use of the multiple AIS target list was in the spirit of Rule 7(b) of the COLREGS for determining risk of collision. However, it did not provide additional information that would otherwise be available through radar plotting, such as radar target compass bearing, and the course and speed of other vessels. It also detracted from using long-range scanning to obtain early warning of risk of collision.

CMA CGM's SMS required AIS and radar to be used for the purpose of fully appraising the navigational situation and the risk of collision. Additionally, *CMA CGM Florida*'s master's standing orders required the OOW to ascertain risk of collision by compass bearings in addition to ARPA plotting.

The SMS did not refer to, or reflect the guidance contained in MGN 324 (M+F) with respect to the use of AIS for collision avoidance. Although the master's standing orders to some extent reflected best practice for ascertaining risk of collision, no instructions were provided as to the manner and extent to which AIS should be used.

2.9.2 Guidance

The ICS's Bridge Procedures Guide warns that increasingly sophisticated navigational aids, now including AIS, bring their own dangers, and that there is a need for properly formulated bridge procedures to counter undue reliance on technology.

The MCA's MGN 324 (M+F) states that there is no provision in the COLREGS for the use of AIS information and therefore recommends that decisions should be taken primarily on visual and/or radar information.

MGN 324 (M+F) includes extracts from IMO Resolution A.917(22), as amended by Resolution A.956(23), with regard to AIS operation on board ship. The extract pertaining to the use of AIS in collision avoidance states that the potential of AIS as an anti-collision device is recognised and it may be recommended as such a device in due time. Meanwhile, it warns that AIS is an additional source of navigational information and does not replace, but supports, navigational systems such as radar target tracking.

MGN 324 (M+F) also includes an extract from the MCA's guidance on implementing the International Convention on the Safety of Life at Sea's safety of navigation requirements. The guidance states that *until the optimum display modes have been fully evaluated and decided upon internationally, AIS will comprise stand-alone units without integration to other displays*.

AIS was fully integrated into the radars fitted on board *CMA CGM Florida*. MGN 324 (M+F) therefore no longer reflects the current situation and provides no guidance on the use of AIS-integrated radar displays.

IMO Resolution MSC.192(79) contains revised international performance standards for radar equipment, which states that the integration and display of AIS information should be provided to complement radar. It further states that target information may be provided by the radar target tracking function and by the reported target information from the AIS, and that, as a default condition, AIS target data should be automatically selected and displayed.

The Filipino OOW's use of AIS and radar was not in accordance with current UK national guidance. However, since the introduction of AIS-integrated radar displays and revised performance standards for radar, such guidance is now in need of revision.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. *CMA CGM Florida's* Filipino OOW lacked situational awareness and precautionary thought. [2.4.2, 2.4.3, 2.4.4, 2.4.5, 2.5, 2.9.1]
2. *CMA CGM Florida's* two 2/Os and *Chou Shan's* OOW considered that it was appropriate to use VHF radio for collision avoidance, contrary to the advice provided in ICS's Bridge Procedures Guide and the MCA's MGN 324 (M+F). [2.4.3, 2.5, 2.6.1, 2.8]
3. *CMA CGM Florida's* Chinese 2/O acted autonomously in his role of external communicator, and did not fully translate his VHF radio conversations into English for the Filipino OOW. [2.5]
4. *Chou Shan's* OOW considered that it was appropriate to use VHF radio for negotiating a passing protocol that was contrary to Rule 15 of the COLREGS. [2.6.1, 2.8]
5. *CMA CGM Florida's* and *Chou Shan's* master's standing orders presumed that the OOW and the master had a shared understanding of the circumstances in which the OOW would become sufficiently concerned to warrant their calling the master. [2.6.2, 2.7.1]
6. *CMA CGM Florida's* master's standing orders lacked specific metrics, provided ambiguous instructions on the use of VHF radio for collision avoidance, and contained no instructions with respect to radar operating mode and AIS integration. [2.7.1, 2.8, 2.9.1]
7. CMA CGM's SMS provided no specific instructions regarding the use of VHF radio for collision avoidance and made no mention of this in its general guidance as to what the master's standing orders should include. [2.8]
8. Sincere Navigation Corporation's SMS conflicted with the advice provided in ICS's Bridge Procedures Guide relating to the use of VHF radio for collision avoidance. [2.8]

3.2 OTHER SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT

1. Whether or not the two individuals had direct interpersonal conflicts is unclear, however, they were both hampered in their attempts to communicate by having to converse in a second language. Current research suggests that culturally the Chinese 2/O is likely to have been respectful of the Filipino 2/O's age, experience and authority as OOW. Other research has also identified concerns over the compatibility of the two nationalities. [2.5]
2. *Chou Shan's* OOW was probably influenced by the high power-distance hierarchy of his national culture in agreeing to a manoeuvre about which he had concerns. [2.6.2]

3.3 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. On *CMA CGM Florida*, three documents containing bridge instructions and an SMS procedure all contained variations on requirements for calling the master, leading to potential confusion. [2.7.1]
2. The potential existed for *CMA CGM Florida's* officers, particularly those of Chinese and Filipino nationalities, to have been negatively influenced by the master's relaxed management style. [2.7.1]
3. *CMA CGM Florida's* master's confidence in the value of following the collision checklist would have been enhanced had he been more familiar with it. [2.7.2]
4. Most emergency drills conducted on *CMA CGM Florida* did not provide for a practical simulation of actions required by the muster list and the relevant emergency checklist, contrary to CMA CGM's SMS requirement for emergency drills to be as realistic as possible. [2.7.2]
5. In using AIS priority and the multiple AIS target list on the port radar, *CMA CGM Florida's* Filipino OOW was encouraged to focus on those vessels with the smallest CPA or range at the expense of maintaining a more strategic overview of the traffic situation. [2.9.1]
6. The MCA's MGN 324 (M+F) no longer reflects the equipment currently fitted to vessels and is in need of revision to include guidance on the use of AIS-integrated radar displays. [2.9.2]